Co-Design of Interchiplet, Package, and System Interconnect Protocols

Tony Chan Carusone and Dustin Dunwell, *Alphawave Semi, Toronto, ON, Canada*
Sundeep Gupta, *Alphawave Semi, Pune, India*
Letizia Giuliano, *Alphawave Semi, OR, USA*
Adrien Auge, *Alphawave Semi, Vancouver, BC, Canada*
Michael Klempa, *Alphawave Semi, NH, USA*
Sue Hung Fung, *Alphawave Semi, San Jose, CA, USA*

Artificial intelligence (AI) applications are adopting modular tile design paradigms for compute and networking chiplets. The integration of UCIe as a die-to-die interconnect for low latency and power savings allows a disaggregation of large monolithic dies. This paper covers interchiplet architectures that can be built toward a chiplet portfolio for escalating bandwidth needs in networking and AI/machine learning. The electrical layer of UCIe design verification is illustrated. In addition, the topic of package design and challenges associated with the integration of the physical layer, in both standard and advanced packaging, are discussed.

UCIe—Universal Chiplet Interconnect Express—addresses the need for a composable architecture through more customizable, package-level integration combined with best-in-class die-to-die interconnect and protocol connections from an interoperable, multivendor ecosystem. Both copackaged optics and off-package electrical links connecting front panel pluggables benefit from retimers utilizing UCIe. By providing a high bandwidth interface between chiplets, UCIe allows prevalidated silicon intellectual property (IP) for electrical or optical links to be copackaged alongside advanced computing and networking. UCIe retimers have discrete advantages in addressing the upcoming landscape of input–output (I/O), in both disaggregated artificial intelligence/machine learning (AI/ML) and siloed networks because UCIe allows for the many permutations addressing different types of bottleneck; there is no one-size-fits-all solution.

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Chiplets, with their modular design and ability to be customized for specific tasks, are well-suited to meet the demands of I/O because they address critical issues such as energy consumption, die area, and yield. However, they don't address some practical issues such as design effort, design risk, and test time. This is the inherent benefit of the standardized UCIe specification. By democratizing chiplets with a common interface, the ecosystem of UCIe accelerates time to market with prevalidated designs, opening innovation to anyone with access to the catalog of solutions. UCIe's ecosystem allows designers to leverage this new in-package connectivity, using silicon-proven IP technologies—High Bandwidth Memory (HBM), Double Data Rate Memory (DDR), Peripheral Component Interconnect Express (PCIe), Compute Express Link (CXL), and Ethernet, for example—and process nodes for what they are best suited. Less time is spent (wasted, really) reinventing the proverbial wheel for development, testing, failed tape outs, and so on, and more meaningful time is spent delivering new and exciting technology to customers, expediting the next generation of compute.

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INTERCHIPLET ARCHITECTURES

The disaggregation of monolithic dies into chiplets enables different process nodes, analog, and digital to be segregated for specific chiplet applications. High-speed interface IP can be reused on chiplets for connectivity, bandwidth, or capacity expansion. Using an extensive portfolio of IP allows the integration of multiple chiplet architectures for I/O chiplets, memory chiplets, and accelerator or compute chiplets (Figure 1).

An I/O chiplet portfolio can include Ethernet to meet escalating demands in high-speed networking. The integration of PCIe and CXL, a native part of the UCIe protocol stack, enables high-speed interconnect peripherals or network interface cards for high-performance compute and data-intensive systems. Memory chiplets allow for expansion beyond the memory directly connected to the CPU. DDR can be added for extra capacity, or HBM can be added for additional bandwidth capabilities. Cache coherency between the CPUs and accelerator chiplets can be added along with other security features such as data encryption and secure boot for enhancement of the chiplet architecture.

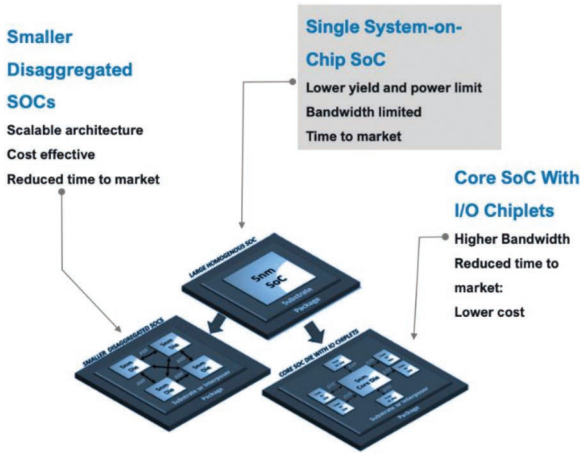

FIGURE 1. Emergence of chiplets paradigm.

UCIe DESIGN VERIFICATION

The UCIe electrical layer needs verification to ensure signaling can reach the opposite die with minimal signal impairment, resulting in minimal bit errors. Standard compliance can be verified by looking at the voltage transfer function of the entire package substrate and compared against the loss and crosstalk requirement defined in the standard (Figure 2).

Additionally, with the help of behavioral I/O models representing Alphawave Semi's UCIe PHY, a detailed end-to-end simulation can be performed to verify the link performance at the data sampler against the bit error rate target performance (Figure 3).

PACKAGE DESIGN AND CHALLENGES

Package design flow is a key aspect of chiplet design and integrating the PHY design with the package. The integration of various components of the PHY such as bump pitch, channel length, power, and shoreline density provide opportunity for varying designs tailored to different application needs. From standard to advanced packaging, each option provides unique advantages and challenges.

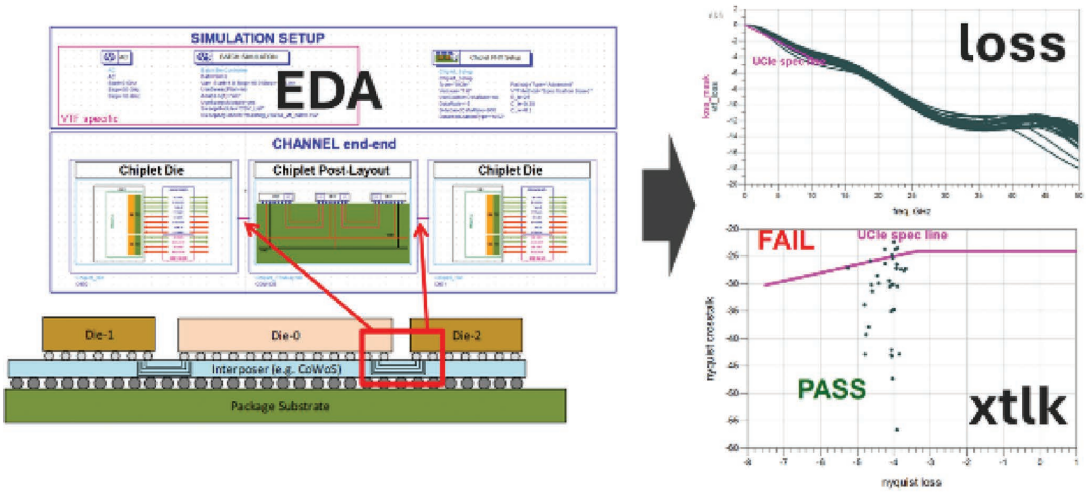

FIGURE 2. UCIe electrical layer verification.[3] Universal chiplet interconnect express—UCIe Specification, Revision 1.1, Version 1.0 (2023).[1]

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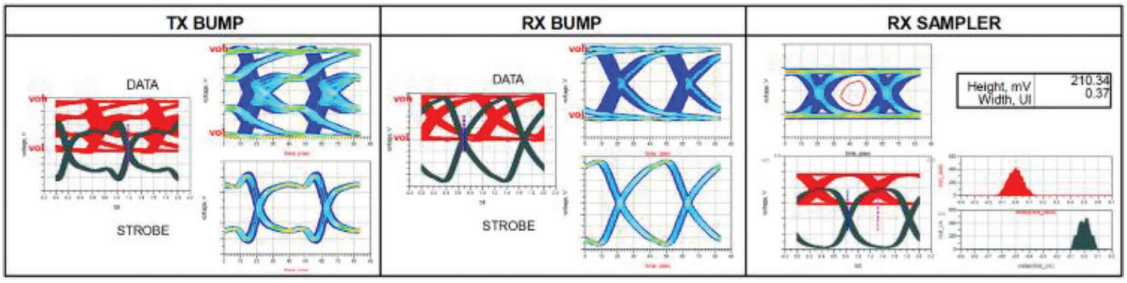

FIGURE 3. UCIe end-to-end link performance.[3]

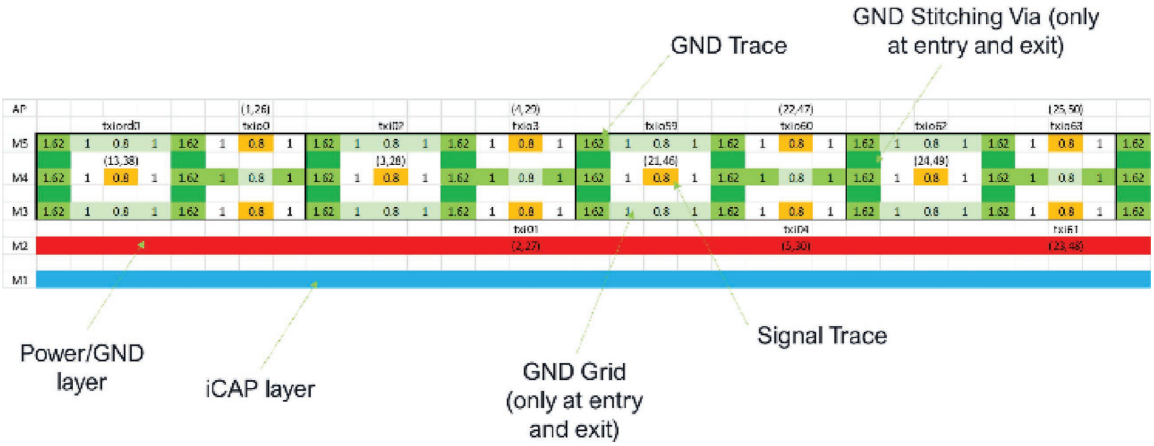

FIGURE 4. UCIe advanced package cross section of channel routing, by I. Ilamparidhi et al.[2]

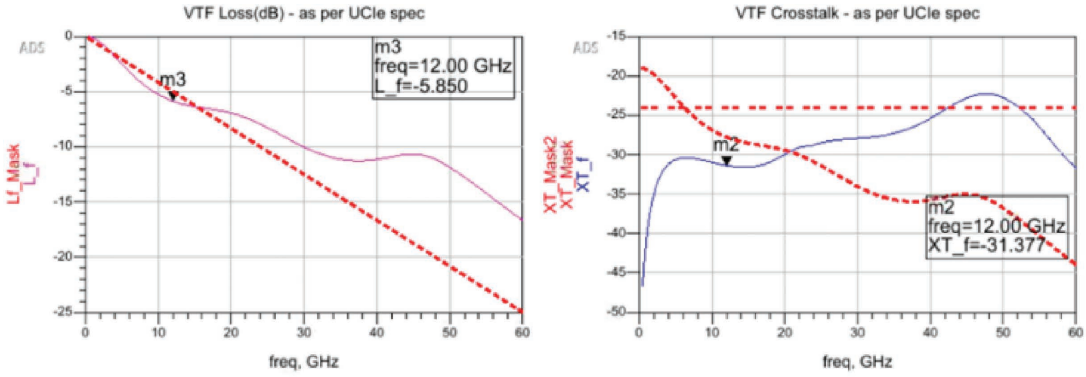

FIGURE 5. Voltage transfer function (VTF) loss and crosstalk, by I. Ilamparidhi et al.[2]

As systems increasingly adopt UCIe and HBM for high-bandwidth computing needs, advanced packaging solutions become a critical part of the solution. Advanced packaging over silicon interposer, integrated fanout, redistribution layer, and embedded bridge solutions enable extremely dense die-to-die interconnects due to tight bump pitches, resulting in higher bandwidth density, lower power, and short-reach 2-mm channels. To mitigate the high cost of advanced packaging, standard packaging offers a low-cost solution with longer reaches up to 25 mm.

In both advanced and standard package technologies, the drive toward higher routing density leads inevitably to crosstalk challenges, as single-ended

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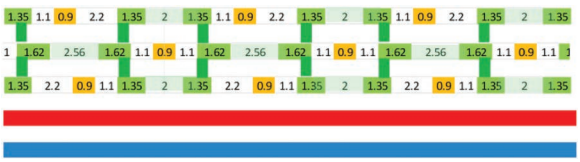

FIGURE 6. UCIe advanced package cross section of channel routing after iterations to improve loss, by I. Ilamparidhi et al.[2]

signals get pushed closer together through package routes. In advanced package technologies, where routing lengths are short and metal width and spacing rules allow, ground shields can be added between adjacent routes to mitigate crosstalk. But the tradeoff between channel performance and power integrity must also be considered, as supply and ground network impedance can have a significant impact on circuit performance and link margin. Figure 4 shows an example cross section of our patented channel routing for a UCIe advanced package design. In this example, two of the available five metal layers are consumed by power and ground routing and by integrated capacitance (iCAP) to help improve power integrity. Metal width and spacing values (in micrometers) show how tight the routing density needs to be to escape all advanced package signals while maintaining good shielding between lines.

As shown in Figure 5, despite meeting the UCIe standard requirements (dashed red lines) for crosstalk, the aforementioned cross section introduces too much loss and fails to meet the standard requirements for insertion loss for 24-Gb/s operation.

By adjusting the widths and spacings of traces on some layers in order to offset signals so that they are not in the same vertical plan, it is possible to increase the width of the signal traces without significantly increasing crosstalk. This lowers the insertion loss and makes it possible to meet the UCIe specification requirements. Figures 6 and 7 show the signal layer cross section resulting from these iterations, and the final loss and crosstalk that were achieved.

There are minimum and maximum design rule dimensions constraining the distances and size features applicable for advanced and standard package routing. Although standard package routing requires much larger spacing between traces, crosstalk remains a challenge since organic substrate width and spacing rules do not allow for shields to be placed between adjacent signal lines. This is exacerbated by the fact that the standard package x32 bump map requires tall vertical via stacks to drop high-speed signal traces down to layer 7 or 8, as shown in the cross section in Figure 8.

These vertical stacks alone introduce a significant amount of skew between channels on different layers and can introduce enough crosstalk to violate the UCIe standard requirements even in the absence of long horizontal traces. Figure 9 illustrates how one iteration focused on increasing the space between traces on a single package layer to minimize crosstalk. While fanning traces out like this between dies may be possible in some applications, it would create collisions between adjacent links in scenarios where multiple UCIe connections are tiled along the edge of the die. However, these experiments yielded only marginal improvements in crosstalk since they do not address the proximity of signals descending through the vertical via stack, as mentioned previously.

The plots in Figure 10 show that after many iterations of the design, it is achievable but remains a challenge to meet the standard requirements for both insertion loss (IL) and crosstalk across all channels when constrained to the UCIe beachfront dimensions.

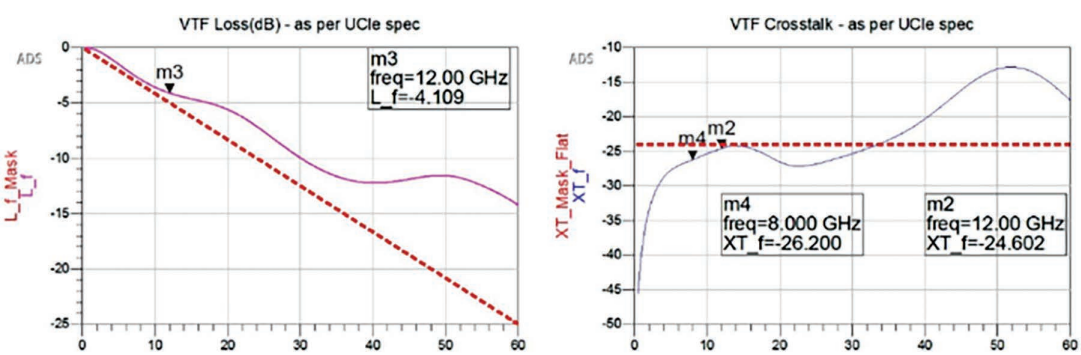

FIGURE 7. VTF loss and crosstalk after iterations to improve loss, by I. Ilamparidhi et al.[2]

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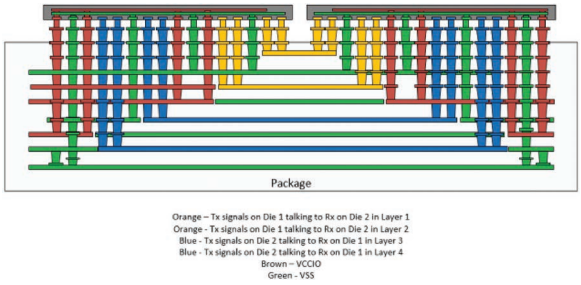

FIGURE 8. Universal chiplet interconnect express—UCIe Specification, Revision 1.1, Version 1.0 (2023).[1]

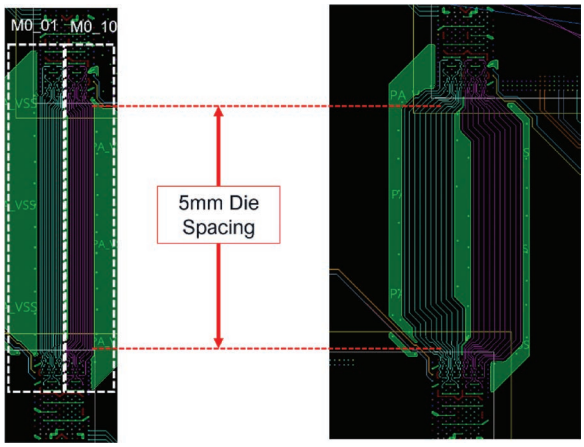

FIGURE 9. Increasing spacing between adjacent signal lines on a single layer has only a marginal impact on overall crosstalk performance, which is dominated by the vertical transition through the via stack, by Xunjun Mo and Dan Lambalot, 2024, Alphawave Semi.

CONCLUSION

AI applications are accelerating and motivating the adoption of chiplet design paradigms in ways that are obvious and subtle. Chiplet-based devices offer significant cost (up to 40%) and power savings (25%–50%) in AI applications. Moreover, the disaggregation of large monolithic compute and networking chips into smaller, modular tiles is the only way to provide the ongoing performance improvements demanded by AI. Connectivity interface IP for storage and networking on dedicated I/O tiles implemented in older generations of fabrication technology can be combined with compute tiles in the most advanced and energy-efficient CMOS technology available. This approach allows us to customize each system according to the target workload by flexibly combining I/O tiles, HBM memory, and compute power. The result is a more efficient, adaptable, and cost-effective computing ecosystem.

This article focused particularly on the essential low-latency die-to-die interfaces required to enable the proliferation of chiplet technology. UCIe has emerged as the preferred interconnect technology for an open chiplet ecosystem. Its flexibility allows for implementation on standard package substrates offering low-cost options. Using advanced packaging technologies offers a clear path to 10+ Tbps/mm of bandwidth density. Die-to-die interfaces continue to attract significant ongoing research interest so we can expect new approaches and further progress. The challenges are multidimensional spanning protocol and architecture design for low latency, circuit design for low power, and package design techniques

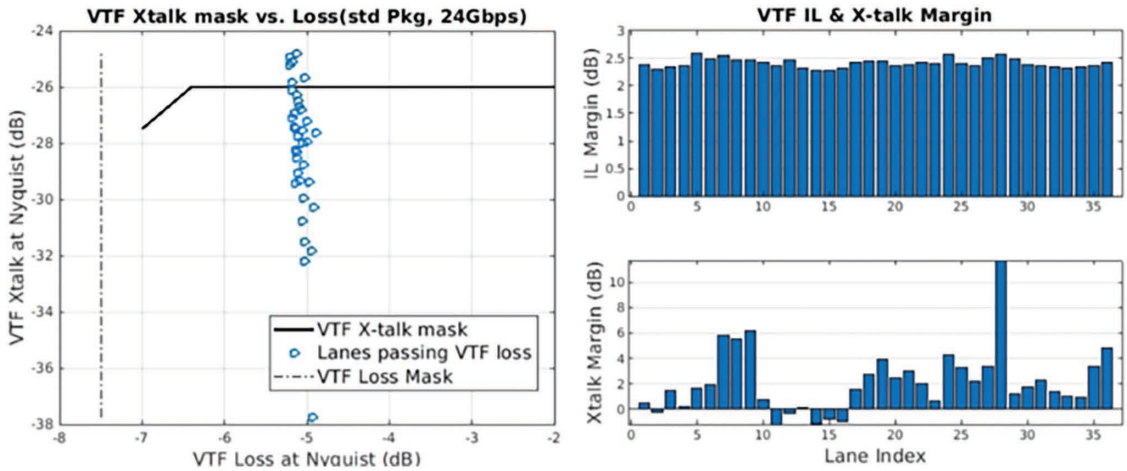

FIGURE 10. Insertion loss and crosstalk after package design optimizations, by Xunjun Mo and Dan Lambalot, 2024, Alphawave Semi.

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to enable high bandwidth density while maintaining power and signal integrity. Solving these problems will allow chiplet technologies to scale to millions of processing and networking units deployed across data centers worldwide, ensuring both affordability and environmental sustainability.

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TONY CHAN CARUSONE is a chief technology officer with Alphawave Semi, Toronto, ON, Canada. Contact him at tonycc@awavesemi.com.

DUSTIN DUNWELL is a principal analog engineering manager with Alphawave Semi, Toronto, ON, Canada. Contact him at dustin.dunwell@awavesemi.com.

SUNDEEP GUPTA is a vice president of IP engineering with Alphawave Semi, Pune, India. Contact him at sundeep.gupta@awavesemi.com.

LETIZIA GIULIANO is a vice president of IP Product Marketing with Alphawave Semi, OR, USA. Contact her at letizia.giuliano@awavesemi.com.

ADRIEN AUGE is a principal applications engineer with Alphawave Semi, Vancouver, BC, Canada. Contact him at adrien.auge@awavesemi.com.

MICHAEL KLEMPA is a staff engineer with Alphawave Semi, NH, USA. Contact him at michael.klempa@awavesemi.com.

SUE HUNG FUNG is a principal product line manager with Alphawave Semi, San Jose, CA, USA. Contact her at sue.fung@awavesemi.com.

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